

Field	Value	Source
Rated capacity (Ah)	3.0 (design rating) / 2.6683 simulation-verified at 1C	bridge/params_v8.json / cell/r4_v8_1c_dfn.json
Nominal voltage / voltage window (V)	midpoint 3.9215; window 2.5 - 4.2	cell/r4_v8_energy.json / Chen2020 cut-offs
Rated energy (Wh)	9.833	cell/r4_v8_energy.json:energy_wh
Energy density	311.0 Wh/kg / 651.3 Wh/L	cell/r4_v8_energy.json (contract mass formula, electrolyte excluded)
Maximum continuous discharge rate	5C (15 A): capacity 2.6261 Ah, retention 0.9842 vs 1C	cell/r4_v8_5c_dfn.json / cell/r4_v8_derived.json
Fast-charge capability	4C (12 A) at 45 C ambient: T_max 323.57 K, anode potential min +0.0119 V -> no plating	cell/r4_v8_4c45_dfn.json
Operating temperature range	Simulated: 298.15 K discharge / 318.15 K charge ambient (protocol conditions); broader range not simulated	protocol definitions
Cycle life	Not simulated (requires aging model)	honest annotation per protocol
Safety determination	No plating on 4C charge; T_max 323.57 K <= 333.15 K (60 C)	cell/r4_v8_5c_dfn.json / cell/r4_v8_4c45_dfn.json
Power density (W/kg)	73121.8	cell/r4_v8_energy.json
Dimensions and mass	electrodes 65 x 1580 mm (unrolled), stack 0.147 mm; mass 31.6 g (contract, electrolyte excluded; 37.6 g incl. electrolyte fill); shell Not provided	Chen2020 + params_v8.json + cell/r4_v8_energy.json